

14. Tangent measures and densities

In this chapter we introduce tangent measures for a Radon measure μ on $\mathbf{R}^n$ as an effective tool for studying the local structure of μ . In this sense they were introduced by Preiss [4], although similar concepts have been extensively used in geometric measure theory, see Federer [3, 4.3.16] and L. Simon [1, § 42]. Taking tangent measures for a measure is a bit like computing the derivative of a function, and like derivatives the tangent measures carry information about the local behaviour of the measure. In this chapter we shall apply the tangent measures to prove the following theorem of Marstrand [4]: if s is a positive number and if there exists a Borel measure μ on $\mathbf{R}^n$ such that the positive and finite density

$$0 < \Theta^s(\mu, x) = \lim_{r \downarrow 0} (2r)^{-s} \mu(B(x, r)) < \infty$$

exists in a set of positive μ measure, then s must be an integer. Note that the essential assumption here is the existence of the limit, that is, $\Theta_*^s(\mu, x) = \Theta^{*s}(\mu, x)$. For an easy proof in the case $0 < s < 1$, see Falconer [4, § 4.2]. Recall Chapter 6 for the definition and some properties of the densities.

Definitions and examples

The tangent measures of μ at $a \in \mathbf{R}^n$ are defined by blowing up μ by sequences of expansive homotheties around a , normalizing suitably and taking weak limits. The map $T_{a,r}$ that blows up $B(a, r)$ to $B(0, 1)$ is given by

$$T_{a,r}(x) = (x - a)/r.$$

Note that the image of μ under $T_{a,r}$ is

$$T_{a,r\#}\mu(A) = \mu(rA + a), \quad A \subset \mathbf{R}^n.$$

For the following definition recall the weak convergence of measures from Chapter 1.

14.1. Definition. Let μ be a Radon measure on $\mathbf{R}^n$. We say that ν is a tangent measure of μ at a point $a \in \mathbf{R}^n$ if ν is a non-zero Radon measure on $\mathbf{R}^n$ and if there exist sequences (r_i) and (c_i) of positive numbers such that $r_i \downarrow 0$ and

$$c_i T_{a,r_i\#}\mu \xrightarrow{w} \nu \quad \text{as } i \rightarrow \infty.$$

The set of all such tangent measures is denoted by $\text{Tan}(\mu, a)$.

According to the definition of weak convergence 1.21 and Theorem 1.19 the above means that for $\varphi \in C_0(\mathbf{R}^n)$

$$\lim_{i \rightarrow \infty} c_i \int \varphi((x-a)/r_i) d\mu x = \int \varphi d\nu.$$

Note that even if μ should have compact support all tangent measures of μ may have unbounded support; see the examples below.

14.2. Examples. (1) If A is an $\mathcal{L}^n$ measurable subset of $\mathbf{R}^n$ and $\mu = \mathcal{L}^n \llcorner A$, it follows from the density theorem 2.14(1) that for μ almost all $a \in \mathbf{R}^n$ (that is, for $\mathcal{L}^n$ almost all $a \in A$), $\text{Tan}(\mu, a) = \{c\mathcal{L}^n : 0 < c < \infty\}$. More generally, the same is true if $\mu(A) = \int_A f d\mathcal{L}^n$ for some positive $\mathcal{L}^n$ integrable function f , see Lemmas 14.5 and 14.6 below.

(2) Let Γ be a rectifiable curve in $\mathbf{R}^n$. Then Γ has a tangent at $\mathcal{H}^1$ almost all of its points. Let $L(a)$ be the line through 0 in $\mathbf{R}^n$ which is parallel to the tangent of Γ at $a \in \Gamma$. If $\mu = \mathcal{H}^1 \llcorner \Gamma$, the length measure on Γ , then $\text{Tan}(\mu, a) = \{c\mathcal{H}^1 \llcorner L(a) : 0 < c < \infty\}$ for $\mathcal{H}^1$ almost all $a \in \Gamma$. We shall discuss this situation more generally in connection with rectifiable sets in Chapter 16.

(3) In the above examples the tangent measures at almost all points were unique up to the multiplication by a constant. In general $\text{Tan}(\mu, a)$ can be very rich, containing many different kinds of measures. In fact, O'Neil [1] has constructed a Radon measure μ on $\mathbf{R}^n$ which has μ almost everywhere all non-zero Radon measures of $\mathbf{R}^n$ as its tangent measures.

Let us consider the construction suggested by Figure 14.1.

We require that the discs D_i are equally spread, they touch the boundary circle of D and $\sum_i d(D_i) = d(D)$. Starting from the unit disc D and performing this operation we can construct a Cantor set

$$C = \bigcap_{k=1}^{\infty} \bigcup_{i_1, \dots, i_k} D_{i_1, \dots, i_k}$$

as in 4.12 so that inside each disc $D_{i_1, \dots, i_k}$ of diameter d_k we have m_k discs $D_{i_1, \dots, i_k, j}$, $j = 1, \dots, m_k$, $m_k \geq 3$, of diameter d_{k+1} with $m_k d_{k+1} = d_k$. Then $0 < \mathcal{H}^1(C) < \infty$ and also $0 < \Theta_*^1(C, x) \leq \Theta^{*1}(C, x) < \infty$ for $x \in C$. Let $\mu = \mathcal{H}^1 \llcorner C$. If we keep m_k constant, or only bounded, the tangent measures of μ will be of the form $c\mathcal{H}^1 \llcorner E$ where E is an unbounded fractal set with positive and locally finite $\mathcal{H}^1$ measure. But

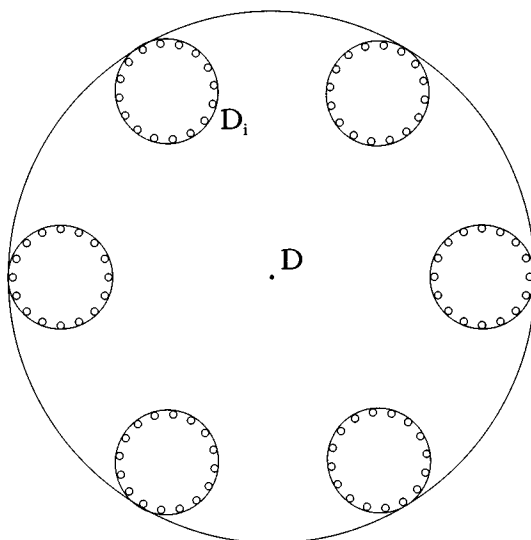


Figure 14.1.

E is not unique at a given point $a \in C$; different sequences $r_i \downarrow 0$ may lead to different sets E . If on the other hand $m_k \rightarrow \infty$, the tangent measures are again of the form $c\mathcal{H}^1 \llcorner E$, but now E is either a line or a countable union of circles. At $\mathcal{H}^1$ almost every point, $\text{Tan}(\mu, a)$ really contains tangent measures of both types and it also contains $c\mathcal{H}^1 \llcorner L$ for all $0 < c < \infty$ and all lines L through 0.

We can also modify this construction letting for long intervals m_k stay bounded and then for other long intervals m_k become very large. Then $\text{Tan}(\mu, a)$ will contain both fractal- and smooth-type tangent measures.

Preliminary results on tangent measures

For any Radon measure μ , $\text{Tan}(\mu, a) \neq \emptyset$ for μ almost all $a \in \mathbf{R}^n$. Although the proof of this is not difficult, see Preiss [4, Theorem 2.5], we shall only give it below in a special case which will suffice for us later on.

14.3. Theorem. *Let μ be a Radon measure on $\mathbf{R}^n$. If $a \in \mathbf{R}^n$ and*

$$(1) \quad c = \limsup_{r \downarrow 0} \frac{\mu(B(a, 2r))}{\mu(B(a, r))} < \infty,$$

then every sequence (r_i) with $r_i \downarrow 0$ contains a sub-sequence (r_{i_j}) such that the measures

$$\mu(B(a, r_{i_j}))^{-1} T_{a, r_{i_j} \#} \mu$$

converge to a tangent measure of μ at a .

Proof. We have for $k = 1, 2, \dots$

$$\begin{aligned} & \limsup_{r \downarrow 0} \mu(B(a, r))^{-1} (T_{a, r \#} \mu)(B(2^k)) \\ &= \limsup_{r \downarrow 0} \mu(B(a, r))^{-1} \mu(B(a, 2^k r)) \leq c^k. \end{aligned}$$

Thus $\sup_i \mu(B(a, r_i))^{-1} (T_{a, r_i \#} \mu)(K) < \infty$ for all compact sets K , and the required convergence follows from Theorem 1.23. $\square$

14.4. Remarks. (1) Even without the doubling condition of Theorem 14.3, we can still choose the normalization constants c_i to be $\mu(B(a, Rr_i))^{-1}$ for any R for which $\nu(U(R)) > 0$ (recall that $U(R)$ and $B(R)$ are the open and closed, respectively, balls with centre 0 and radius R). That is, if $\nu = \lim_{i \rightarrow \infty} c_i T_{a, r_i \#} \mu$, Theorem 1.24 gives

$$\begin{aligned} 0 &< \nu(U(R)) \leq \liminf_{i \rightarrow \infty} c_i \mu(U(a, Rr_i)) \\ &\leq \limsup_{i \rightarrow \infty} c_i \mu(B(a, Rr_i)) \leq \nu(B(R)) < \infty. \end{aligned}$$

Thus the sequence $c_i \mu(B(a, Rr_i))$ has a sub-sequence converging to a positive and finite number c , and then $c \mu(B(a, Rr_{i_j}))^{-1} T_{a, r_{i_j} \#} \mu \xrightarrow{w} \nu$.

The assumption (1) in Theorem 14.3 has the following two immediate consequences.

(2) $0 \in \text{spt } \nu$ for all $\nu \in \text{Tan}(\mu, a)$.

In fact, if, recalling (1), $\mu(B(a, Rr_i))^{-1} T_{a, r_i \#} \mu$ converges to $c\nu$, we have by Theorem 1.24 (1) for all $r > 0$,

$$\begin{aligned} c\nu(B(r)) &\geq \limsup_{i \rightarrow \infty} \mu(B(a, Rr_i))^{-1} (T_{a, r_i \#} \mu)(B(r)) \\ &= \limsup_{i \rightarrow \infty} \mu(B(a, Rr_i))^{-1} \mu(B(a, rr_i)) > 0 \end{aligned}$$

by 14.3 (1).

(3) For every $\nu \in \text{Tan}(\mu, a)$ there are a sequence $r_i \downarrow 0$ and a positive number c such that

$$\nu = c \lim_{i \rightarrow \infty} \mu(B(a, r_i))^{-1} T_{a, r_i \#} \mu,$$

provided 14.3(1) holds. Since $0 \in \text{spt } \nu$, this follows with the argument of (1). Moreover, if for some s ,

$$0 < \Theta_*^s(\mu, a) \leq \Theta^{*s}(\mu, a) < \infty,$$

we can find the sequence (r_i) so that

$$\nu = d \lim_{i \rightarrow \infty} r_i^{-s} T_{a, r_i \#} \mu$$

for some positive number d .

This follows by choosing a sub-sequence (r_{i_j}) for which $r_{i_j}^{-s} \mu(B(a, r_{i_j}))$ converges.

14.5. Lemma. *Let μ be a Radon measure on $\mathbf{R}^n$ and B a μ measurable set. If $a \in \text{spt } \mu$ and*

$$\lim_{r \downarrow 0} \frac{\mu(B(a, r) \setminus B)}{\mu(B(a, r))} = 0,$$

then $\text{Tan}(\mu \llcorner B, a) = \text{Tan}(\mu, a)$. In particular this holds for μ almost all $a \in B$.

Proof. Let $r_i \downarrow 0$ and $c_i > 0$ be such that at least one of the sequences $c_i T_{a, r_i \#} \mu$ and $c_i T_{a, r_i \#} (\mu \llcorner B)$ converges weakly to a locally finite Radon measure ν . As in Remark 14.4(1) we have

$$\limsup_{i \rightarrow \infty} c_i \mu(B \cap B(a, Rr_i)) \leq \nu(B(R)) < \infty.$$

Let $\varphi \in C_0(\mathbf{R}^n)$. Choose $R < \infty$ such that $|\varphi| < R$ and $\text{spt } \varphi \subset B(R)$. Then for $r > 0$,

$$\begin{aligned} & \left| \int \varphi dT_{a, r \#} \mu - \int \varphi dT_{a, r \#} (\mu \llcorner B) \right| \\ &= \left| \int_{\mathbf{R}^n \setminus B} \varphi((x - a)/r) d\mu x \right| \leq R \mu(B(a, Rr) \setminus B). \end{aligned}$$

Thus we obtain

$$\begin{aligned} & \limsup_{i \rightarrow \infty} \left| \int \varphi d(c_i T_{a, r_i \#} \mu) - \int \varphi d(c_i T_{a, r_i \#} (\mu \llcorner B)) \right| \\ & \leq R \nu(B(R)) \limsup_{i \rightarrow \infty} \frac{\mu(B(a, Rr_i) \setminus B)}{\mu(B \cap B(a, Rr_i))} = 0. \end{aligned}$$

It follows that both of the sequences $c_i T_{a, r_i \#} \mu$ and $c_i T_{a, r_i \#} (\mu \llcorner B)$ converge weakly to ν . Hence μ and $\mu \llcorner B$ have the same tangent measures at a .

The last statement follows from the density theorem 2.14(1). $\square$

14.6. Lemma. Let μ be a Radon measure on $\mathbf{R}^n$, φ a non-negative locally μ integrable function on $\mathbf{R}^n$, and λ the Radon measure such that $\lambda(B) = \int_B \varphi d\mu$ for Borel sets B . Then $\text{Tan}(\mu, a) = \text{Tan}(\lambda, a)$ for λ almost all $a \in \mathbf{R}^n$.

Proof. Let $A = \{x : \varphi(x) > 0\}$ and $\varepsilon > 0$. By Lusin's theorem there is a closed set $F \subset A$ such that $\lambda(\mathbf{R}^n \setminus F) < \varepsilon$ and $\varphi|_F$ is continuous. One easily checks that $\text{Tan}(\mu \llcorner F, a) = \text{Tan}(\lambda \llcorner F, a)$ for λ almost all $a \in F$. Hence by Lemma 14.5, $\text{Tan}(\mu, a) = \text{Tan}(\lambda, a)$ for λ almost all $a \in F$, and the lemma follows. $\square$

Densities and tangent measures

According to the following lemma inequalities for the densities of a measure are turned into uniform estimates for the tangent measures.

14.7. Lemma. Let s be a positive number, μ a Radon measure on $\mathbf{R}^n$ and A the set of the points $a \in \mathbf{R}^n$ such that

$$0 < \Theta_*^s(\mu, a) \leq \Theta^{*s}(\mu, a) < \infty.$$

The following three statements (1)–(3) hold at μ almost all points $a \in A$.

(1) For every $\nu \in \text{Tan}(\mu, a)$ there is a positive number c such that

$$tcr^s \leq \nu(B(x, r)) \leq cr^s \quad \text{for } x \in \text{spt } \nu, \quad 0 < r < \infty,$$

where $t = t(a) = \Theta_*^s(\mu, a)/\Theta^{*s}(\mu, a)$.

(2) If also for μ almost all $z \in A$,

$$\limsup_{\delta \downarrow 0} \{d(B)^{-s} \mu(B) : B \text{ is a closed ball with } z \in B \text{ and } d(B) < \delta\} \leq \Theta^{*s}(\mu, z),$$

then every $\nu \in \text{Tan}(\mu, a)$ satisfies

$$\nu(B(x, r)) \leq cr^s \quad \text{for } x \in \mathbf{R}^n, \quad r > 0,$$

with the same constant c as in (1).

(3) If $s < n$, there are $e \in S^{n-1}$ and $\nu \in \text{Tan}(\mu, a)$ such that

$$\text{spt } \nu \subset \{x \in \mathbf{R}^n : x \cdot e \geq 0\}.$$

Moreover, the following uniform estimate holds.

(4) If there exist positive numbers d , t and r_0 such that

$$tdr^s \leq \mu(B(a, r)) \leq dr^s \quad \text{for } a \in \text{spt } \mu, \quad 0 < r < r_0,$$

then at every point $a \in \text{spt } \mu$, $\text{Tan}(\mu, a) \neq \emptyset$ and all tangent measures $\nu \in \text{Tan}(\mu, a)$ satisfy (1).

Proof. (1) We can write μ almost all of A as the union of Borel sets $A_1, A_2, \dots$ such that each of the functions $\Theta_*^s(\mu, \cdot)$, $\Theta^{*s}(\mu, \cdot)$ and t oscillate very little in A_i . Decomposing further every A_i into a countable union we can for a given $\varepsilon > 0$ find Borel sets $B_i \subset A$ and positive numbers τ_i , r_i and c_i such that $\mu(A \setminus \bigcup_i B_i) = 0$ and for $a \in B_i$, $0 < r < r_i$,

$$\tau_i \leq t(a) < \tau_i + \varepsilon \quad \text{and} \quad \tau_i c_i r^s \leq \mu(B(a, r)) \leq c_i r^s.$$

It follows that it suffices to verify the following statement.

Let τ , c and r_0 be positive numbers and $B \subset A$ a Borel set such that

$$\tau cr^s \leq \mu(B(a, r)) \leq cr^s \quad \text{for } a \in B, \quad 0 < r < r_0.$$

If a is a μ -density point of B , then $\tau cr^s \leq \nu(B(x, r)) \leq cr^s$ for every $\nu \in \text{Tan}(\mu, a)$, $x \in \text{spt } \nu$ and $0 < r < \infty$ provided

$$\nu = \lim_{i \rightarrow \infty} r_i^{-s} T_{a, r_i \#} \mu$$

for some sequence $r_i \downarrow 0$. For this reduction we also use Corollary 2.14 and Remark 14.4 (3). So suppose $a \in B$ with

$$(4) \quad \lim_{r \downarrow 0} \frac{\mu(B(a, r) \setminus B)}{\mu(B(a, r))} = 0,$$

$\nu \in \text{Tan}(\mu, a)$ is as above, $x \in \text{spt } \nu$ and $0 < r < \infty$. As $x \in \text{spt } \nu$, Theorem 1.24 (2) implies for $\varrho > 0$

$$\begin{aligned} 0 < \nu(U(x, \varrho)) &\leq \liminf_{i \rightarrow \infty} r_i^{-s} (T_{a, r_i \#} \mu)(U(x, \varrho)) \\ &= \liminf_{r \downarrow 0} r_i^{-s} \mu(U(a + r_i x, \varrho r_i)) \\ &= \liminf_{i \rightarrow \infty} r_i^{-s} \mu(B \cap U(a + r_i x, \varrho r_i)). \end{aligned}$$

In particular, for any $\varrho > 0$, $B \cap U(a + r_i x, \varrho r_i) \neq \emptyset$ for all sufficiently large i . Thus we may select a sequence $a_i \in B$ such that $x_i = (a_i - a)/r_i \rightarrow x$. Then as above we obtain for any $r < u < v$

$$\begin{aligned} \nu(B(x, r)) &\leq \nu(U(x, u)) \leq \liminf_{i \rightarrow \infty} r_i^{-s} (T_{a, r_i \# \mu})(U(x, u)) \\ &\leq \liminf_{i \rightarrow \infty} r_i^{-s} (T_{a, r_i \# \mu})(U(x_i, v)) \\ &= \liminf_{i \rightarrow \infty} r_i^{-s} \mu(U(a_i, r_i v)) \leq c v^s. \end{aligned}$$

Letting $v \downarrow r$ we get $\nu(B(x, r)) \leq c r^s$. The lower bound is derived in the same way.

The statement (2) can be verified by a similar argument.

(3) Let $a \in \mathbf{R}^n$ be such that (1) holds. Since $s < n$, $\text{spt } \nu \neq \mathbf{R}^n$, cf. Exercise 4. Pick $z \in \mathbf{R}^n \setminus \text{spt } \nu$ and let $\varrho > 0$ be such that $U(z, \varrho) \cap \text{spt } \nu = \emptyset$ and $B(z, \varrho) \cap \text{spt } \nu$ contains some point y . By Theorem 14.3 there is $\lambda \in \text{Tan}(\nu, y)$. One easily verifies that $\text{spt } \lambda$ is contained in the half-space $\{x : x \cdot e \geq 0\}$ where $e = (y - z)/|y - z|$; see the end of the proof of Theorem 14.10.

It remains to show that $\lambda \in \text{Tan}(\mu, a)$. We postpone this to 14.16. However, for the proof of Theorem 14.10, where this statement will be applied, the weaker information that $0 \in \text{spt } \lambda \subset \{x : x \cdot e \geq 0\}$ and that the conclusion of Lemma 14.7 (1) holds for λ in place of ν would suffice.

(4) follows by an inspection of the proof of (1). $\square$

s-uniform measures

If in the preceding lemma the density $\Theta^s(\mu, a)$ exists μ almost everywhere, the tangent measures ν will possess the property of the next definition.

14.8. Definition. Let s be a positive number. A non-zero Radon measure ν is called *s-uniform* if there exists a positive number c such that

$$0 < \nu(B(x, r)) = c r^s < \infty \quad \text{for } x \in \text{spt } \nu \text{ and } 0 < r < \infty.$$

Of course, every *s-uniform* measure is uniformly distributed on the metric space $\text{spt } \nu$ in the sense of Definition 3.3.

14.9. Corollary (to Lemma 14.7). Let s be a positive number, μ a Radon measure on $\mathbf{R}^n$ and A the set of the points $a \in \mathbf{R}^n$ such that the density $\Theta^s(\mu, a)$ exists and is positive and finite. Then for μ almost all $a \in A$ every $\nu \in \text{Tan}(\mu, a)$ is *s-uniform* with $0 \in \text{spt } \nu$.

The last statement, $0 \in \text{spt } \nu$, is contained in Remark 14.4 (2).

Marstrand's theorem

Remember that our goal is to show that if there exists a measure μ as in Corollary 14.9 with $\mu(A) > 0$, then s must be an integer. Since by Theorem 14.3 μ has tangent measures at μ almost all points, there exist then by Corollary 14.9 s -uniform measures. Thus we have simplified the problem; we have to show that the existence of an s -uniform measure on $\mathbf{R}^n$ forces s to be an integer. We shall now prove this.

14.10. Theorem. *Let s be a positive number. Suppose that there exists a Radon measure μ on $\mathbf{R}^n$ such that the density $\Theta^s(\mu, a)$ exists and is positive and finite in a set of positive μ measure. Then s is an integer.*

Proof. If the theorem is false, then, recalling the above remarks, there exists an s -uniform Radon measure in some $\mathbf{R}^n$ for some non-integral s . Let n be the smallest dimension where this can happen. Then $n > 0$. We shall derive a contradiction by finding an s -uniform Radon measure in $\mathbf{R}^{n-1}$ (with $\mathbf{R}^0 = \{0\}$).

Using Lemma 14.7 we find $e \in S^{n-1}$ and an s -uniform measure ν in $\mathbf{R}^n$ with

$$0 \in \text{spt } \nu \subset \{x \in \mathbf{R}^n : x \cdot e \geq 0\} = H.$$

For $r > 0$ let $b(r)$ be the centre of mass of $\nu \llcorner B(r)$ (recall Exercise 1.7), that is,

$$b(r) \cdot v = \nu(B(r))^{-1} \int_{B(r)} z \cdot v \, d\nu z \quad \text{for all } v \in \mathbf{R}^n.$$

If $b(r) = 0$ for all $r > 0$, we have, as $\text{spt } \nu \subset H$,

$$0 = \int_{B(r)} x \cdot e \, d\nu x = \int_{H \cap B(r)} x \cdot e \, d\nu x,$$

whence $x \cdot e = 0$ for ν almost all $x \in \mathbf{R}^n$, which means $\text{spt } \nu \subset \partial H$. This contradicts the minimality of n .

Suppose $b(r) \neq 0$ for some $r > 0$. Let $y \in \text{spt } \nu$. Using the identities (see Exercise 5)

$$\int_{B(y,r)} (r^2 - |x - y|^2) \, d\nu x = \int_{B(r)} (r^2 - |x|^2) \, d\nu x$$

and

$$r^2 - |x|^2 - (r^2 - |x - y|^2) = |y|^2 - 2x \cdot y,$$

we estimate

$$\begin{aligned} |2b(r) \cdot y| &= \left| \nu(B(r))^{-1} \int_{B(r)} 2x \cdot y \, d\nu x \right| \\ &= \left| |y|^2 + \nu(B(r))^{-1} \int_{B(r)} (r^2 - |x - y|^2) \, d\nu x \right. \\ &\quad \left. - \nu(B(r))^{-1} \int_{B(r)} (r^2 - |x|^2) \, d\nu x \right| \\ &= \left| |y|^2 + \nu(B(r))^{-1} \left[\int_{B(r)} (r^2 - |x - y|^2) \, d\nu x \right. \right. \\ &\quad \left. \left. - \int_{B(y,r)} (r^2 - |x - y|^2) \, d\nu x \right] \right| \\ &\leq |y|^2 + \nu(B(r))^{-1} \left[\left| \int_{B(r) \setminus B(y,r)} (r^2 - |x - y|^2) \, d\nu x \right| \right. \\ &\quad \left. + \left| \int_{B(y,r) \setminus B(r)} (r^2 - |x - y|^2) \, d\nu x \right| \right]. \end{aligned}$$

Suppose now $|y| \leq r$. If $x \in B(r) \setminus B(y, r)$, we have

$$\begin{aligned} 0 &\leq |x - y|^2 - r^2 \leq |x - y|^2 - |x|^2 \\ &= (|x - y| + |x|)(|x - y| - |x|) \leq 3r|y|. \end{aligned}$$

Similarly we see that for $x \in B(y, r) \setminus B(r)$,

$$0 \leq r^2 - |x - y|^2 \leq |x|^2 - |x - y|^2 \leq 3r|y|.$$

Consequently, as ν is s -uniform,

$$\begin{aligned} |2b(r) \cdot y| &\leq |y|^2 + \nu(B(r))^{-1} 3r|y| [\nu(B(r) \setminus B(y, r)) + \nu(B(y, r) \setminus B(r))] \\ &\leq |y|^2 + \nu(B(r))^{-1} 3r|y| 2\nu[B(r + |y|) \setminus B(r - |y|)] \\ &= |y|^2 + 6r^{1-s}|y| [(r + |y|)^s - (r - |y|)^s] \\ &\leq c(r)|y|^2, \end{aligned}$$

where $c(r)$ is a positive constant depending on r but not on y . Thus we have shown

$$(1) \quad |b(r) \cdot y| \leq c(r)|y|^2$$

for all $y \in \text{spt } \nu \cap B(r) \setminus \{0\}$.

Take once more a tangent measure $\lambda \in \text{Tan}(\nu, 0)$, say $\lambda = \lim_{i \rightarrow \infty} r_i^{-s} T_{0, r_i \#} \nu$. We shall show that

$$(2) \quad \text{spt } \lambda \subset V = \{y \in \mathbf{R}^n : y \cdot b(r) = 0\}.$$

Let $\eta > 0$, $R > 0$ and set

$$G = \{y \in B(R) : |y \cdot b(r)| > \eta|y|\}.$$

Using Theorem 1.24 (2) we obtain

$$\begin{aligned} \lambda(G) &\leq \liminf_{i \rightarrow \infty} r_i^{-s} (T_{0, r_i \#} \nu)(G) \\ &= \liminf_{i \rightarrow \infty} r_i^{-s} \nu(\{y \in B(Rr_i) : |y \cdot b(r)| > \eta|y|\}) = 0, \end{aligned}$$

because by (1) $y \in B(Rr_i) \cap \text{spt } \nu$ implies for large i ,

$$|y \cdot b(r)| \leq c(r)|y|^2 \leq c(r) Rr_i|y| < \eta|y|.$$

Hence $\lambda(G) = 0$ for all $\eta > 0$ and $R > 0$, which gives (2). Since λ is also s -uniform, we obtain again a contradiction with the minimality of n . $\square$

From the proof of the above theorem one also sees the following rather easy fact.

14.11. Theorem. *If ν is an n -uniform measure in $\mathbf{R}^n$, then ν is a constant multiple of $\mathcal{L}^n$.*

Indeed, if $\text{spt } \nu = \mathbf{R}^n$, this follows from Theorem 3.4. Otherwise we can use the above argument to derive a contradiction.

Falconer and Springer [1] used tangent measures to prove a generalization of Theorem 14.10 involving the average densities of 6.14 (3).

A metric on measures

Sometimes it will be convenient to use a metric on the space of measures in connection with tangent measures. We shall first look at the weak convergence in this light.

14.12. Definition. Let $0 < r < \infty$. We denote by $\mathcal{L}(r)$ the set of all Lipschitz functions $f: \mathbf{R}^n \rightarrow [0, \infty)$ with $\text{spt } f \subset B(r)$ and with $\text{Lip}(f) \leq 1$. For Radon measures μ and ν on $\mathbf{R}^n$ set

$$F_r(\mu, \nu) = \sup \left\{ \left| \int f d\mu - \int f d\nu \right| : f \in \mathcal{L}(r) \right\}.$$

It is easy to see that F_r is a metric on the space of Radon measures with support in $U(r)$; see Exercise 6. Clearly, $F_r(\mu_i, \mu) \rightarrow 0$ implies $\int f d\mu_i \rightarrow \int f d\mu$ for all Lipschitz functions $f: \mathbf{R}^n \rightarrow \mathbf{R}$ with $\text{spt } f \subset B(r)$.

14.13. Lemma. Let $\mu_1, \mu_2, \dots$ and μ be Radon measures on $\mathbf{R}^n$. Then $\mu_i \xrightarrow{w} \mu$ if and only if $\lim_{i \rightarrow \infty} F_r(\mu_i, \mu) = 0$ for all $r > 0$.

Proof. Let $\mu_i \xrightarrow{w} \mu$. Suppose that for some $r > 0$, $F_r(\mu_i, \mu)$ does not tend to zero. By passing to a sub-sequence we may then assume that there are $\varepsilon > 0$ and $f_i \in \mathcal{L}(r)$, $i = 1, 2, \dots$, such that

$$(1) \quad \left| \int f_i d\mu_i - \int f_i d\mu \right| \geq \varepsilon.$$

Ascoli's theorem, see e.g. Rudin [2, p. 369], implies that some sub-sequence of (f_i) , which we again may assume to be the whole sequence, converges uniformly to a function $f \in \mathcal{L}(r)$. By Theorem 1.24(1) the sequence $(\mu_i(B(r)))$ is bounded, which yields

$$\lim_{i \rightarrow \infty} \int f_i d\mu_i = \lim_{i \rightarrow \infty} \int f d\mu_i = \int f d\mu = \lim_{i \rightarrow \infty} \int f_i d\mu.$$

This contradicts (1) and proves $F_r(\mu_i, \mu) \rightarrow 0$ for all $r > 0$.

For the converse, assume $\lim_{i \rightarrow \infty} F_r(\mu_i, \mu) = 0$ for all $r > 0$. Let $\varphi \in C_0(\mathbf{R}^n)$. Choose $r > 1$ so that $\text{spt } \varphi \subset B(r-1)$. Then φ can be approximated uniformly with Lipschitz functions whose support lies in $B(r)$; recall Exercise 1.8. So given $\varepsilon > 0$ we can find a Lipschitz function f such that $|f(x) - \varphi(x)| < \varepsilon$ for $x \in \mathbf{R}^n$ and $\text{spt } f \subset B(r)$. Define g by $g(x) = \max\{0, 1 - d(x, B(r))\}$. Then $g \in \mathcal{L}(r+1)$ and $g = 1$ on $B(r)$. Thus

$$(2) \quad \limsup_{i \rightarrow \infty} \mu_i(B(r)) \leq \limsup_{i \rightarrow \infty} \int g d\mu_i = \int g d\mu < \infty.$$

Using (2) we infer

$$\begin{aligned} & \limsup_{i \rightarrow \infty} \left| \int \varphi d\mu_i - \int \varphi d\mu \right| \\ & \leq \limsup_{i \rightarrow \infty} \int |\varphi - f| d\mu_i + \limsup_{i \rightarrow \infty} \left| \int f d\mu_i - \int f d\mu \right| + \int |f - \varphi| d\mu \\ & \leq \left(\int g d\mu + \mu(B(r)) \right) \varepsilon. \end{aligned}$$

Hence $\lim_{i \rightarrow \infty} \int \varphi d\mu_i = \int \varphi d\mu$ and so $\mu_i \xrightarrow{w} \mu$. $\square$

Next we observe that F_r is separable.

14.14. Lemma. *There is a countable dense set $\mathcal{D}$ of Radon measures on $\mathbf{R}^n$, that is, for any Radon measure μ on $\mathbf{R}^n$ and any positive numbers r and ε we can find $\nu \in \mathcal{D}$ for which $F_r(\mu, \nu) < \varepsilon$.*

Proof. It is enough to prove this for a fixed $r = 1, 2, \dots$. Let $Q_{j,1}, \dots, Q_{j,m_j}$ be all the dyadic cubes of side-length 2^{-j} , $j \geq 1$, which meet $B(r)$ (recall 5.2). Denote by $x_{j,i}$ the centre of $Q_{j,i}$. Then the measures

$$\mu_j = \sum_{i=1}^{m_j} \mu(Q_{j,i}) \delta_{x_{j,i}}, \quad j = 1, 2, \dots,$$

satisfy $\lim_{j \rightarrow \infty} F_r(\mu_j, \mu) = 0$ as one easily checks. From this one sees that the family of all measures of the form $\sum_{i=1}^{m_j} q_{j,i} \delta_{x_{j,i}}$, where the $q_{j,i}$'s are positive rationals, has the required approximation property. $\square$

14.15. Remark. For Radon measures μ and ν on $\mathbf{R}^n$ set

$$d(\mu, \nu) = \sum_{i=1}^{\infty} 2^{-i} \min \{1, F_i(\mu, \nu)\}.$$

It is not hard to show that d is a complete and separable metric on the space of all Radon measures on $\mathbf{R}^n$. Moreover, the convergence with respect to d agrees with the weak convergence.

Tangent measures to tangent measures are tangent measures

We now return to the tangent measures and prove a theorem which settles the point that was left open in 14.7 (3).

14.16. Theorem. Let μ be a Radon measure on $\mathbf{R}^n$. Then at μ almost all points $a \in \mathbf{R}^n$ every $\nu \in \text{Tan}(\mu, a)$ has the following two properties:

- (1) $T_{x, \varrho} \# \nu \in \text{Tan}(\mu, a)$ for $x \in \text{spt } \nu$, $\varrho > 0$.
- (2) $\text{Tan}(\nu, x) \subset \text{Tan}(\mu, a)$ for $x \in \text{spt } \nu$.

Proof. (1) Since $T_{x, \varrho} = T_{0, \varrho} \circ T_{x, 1}$ and $T_{0, \varrho}$ is readily seen to preserve tangent measures, we may assume that $\varrho = 1$. For $k, m = 1, 2, \dots$, let $A_{k, m}$ be the set of all $a \in \mathbf{R}^n$ for which there are $\nu_a \in \text{Tan}(\mu, a)$ and $x_a \in \text{spt } \nu_a$ with

$$(3) \quad F_k(T_{x_a, 1} \# \nu_a, cT_{a, r} \# \mu) > 1/k$$

for all $c > 0$ and $0 < r < 1/m$. We show that $\mu(A_{k, m}) = 0$ for all k, m .

Suppose $\mu(A_{k, m}) > 0$ for some k, m . Using Lemma 14.14 we find a set $A \subset A_{k, m}$ such that $\mu(A) > 0$ and

$$(4) \quad F_k(T_{x_a, 1} \# \nu_a, T_{x_b, 1} \# \nu_b) < 1/(2k) \quad \text{for } a, b \in A.$$

By 2.14(1) and 2.15(2) we can find $a \in A$ such that

$$(5) \quad \lim_{r \downarrow 0} \frac{\mu(A \cap B(a, r))}{\mu(B(a, r))} = 1.$$

Let $c_i > 0$ and $r_i \downarrow 0$ be such that $\nu_a = \lim_{i \rightarrow \infty} c_i T_{a, r_i} \# \mu$, and let $a_i \in A$ be such that

$$(6) \quad |a_i - (a + r_i x_a)| < d(a + r_i x_a, A) + r_i/i.$$

We prove that

$$(7) \quad \lim_{i \rightarrow \infty} \frac{d(a + r_i x_a, A)}{r_i} = 0.$$

Assuming that this is not the case, we find δ , $0 < \delta < |x_a|$, such that $d(a + r_i x_a, A) > \delta r_i$ for infinitely many values i . Then by (5) and Theorem 1.24

$$\begin{aligned} 1 &= \lim_{i \rightarrow \infty} \frac{\mu(A \cap B(a, 2r_i |x_a|))}{\mu(B(a, 2r_i |x_a|))} \leq 1 - \liminf_{i \rightarrow \infty} \frac{\mu(B(a + r_i x_a, \delta r_i))}{\mu(B(a, 2r_i |x_a|))} \\ &= 1 - \liminf_{i \rightarrow \infty} \frac{c_i T_{a, r_i} \# \mu(B(x_a, \delta))}{c_i T_{a, r_i} \# \mu(B(2|x_a|))} \leq 1 - \frac{\nu_a(U(x_a, \delta))}{\nu_a(B(2|x_a|))} < 1. \end{aligned}$$

Thus (7) holds and consequently

$$(8) \quad \lim_{i \rightarrow \infty} \frac{|a_i - (a + r_i x_a)|}{r_i} = 0.$$

As $T_{a_i, r_i} = T_{(a_i - a)/r_i, 1} \circ T_{a, r_i}$, (8) yields

$$\begin{aligned} \lim_{i \rightarrow \infty} c_i T_{a_i, r_i} \# \mu &= \lim_{i \rightarrow \infty} T_{(a_i - a)/r_i, 1} \# (c_i T_{a, r_i} \# \mu) \\ &= T_{x_a, 1} \# \nu_a. \end{aligned}$$

Therefore by Lemma 14.13 there is i such that $r_i < 1/m$ and

$$F_k(T_{x_a, 1} \# \nu_a, c_i T_{a_i, r_i} \# \mu) < 1/(2k).$$

This together with (3) and (4) gives

$$\begin{aligned} 1/k &< F_k(T_{x_{a_i}, 1} \# \nu_{a_i}, c_i T_{a_i, r_i} \# \mu) \\ &\leq F_k(T_{x_{a_i}, 1} \# \nu_{a_i}, T_{x_a, 1} \# \nu_a) + F_k(T_{x_a, 1} \# \nu_a, c_i T_{a_i, r_i} \# \mu) \\ &< 1/k. \end{aligned}$$

This contradiction proves $\mu(A_{k,m}) = 0$ for all $k, m = 1, 2, \dots$. It follows now from Lemma 14.13 and the definition of $A_{k,m}$ that (1) holds for all $a \in \mathbf{R}^n \setminus \bigcup_{k,m} A_{k,m}$.

(2) Let $a \in \mathbf{R}^n$ and $\nu \in \text{Tan}(\mu, a)$ be such that (1) holds. Let $x \in \text{spt } \nu$ and $\lambda = \lim_{i \rightarrow \infty} c_i T_{x, r_i} \# \nu \in \text{Tan}(\nu, x)$. For each $k = 1, 2, \dots$ we can choose i_k such that

$$F_k(c_{i_k} T_{x, r_{i_k}} \# \nu, \lambda) < 1/k.$$

By (1), $T_{x, r_{i_k}} \# \nu \in \text{Tan}(\mu, a)$, so there are d_k and ϱ_k such that $0 < \varrho_k < 1/k$ and

$$F_k(c_{i_k} T_{x, r_{i_k}} \# \nu, d_k T_{a, \varrho_k} \# \mu) < 1/k.$$

This gives $\lambda = \lim_{k \rightarrow \infty} d_k T_{a, \varrho_k} \# \mu \in \text{Tan}(\mu, a)$ and completes the proof. $\square$

Proof of Theorem 11.11

We now return to the proof of the statements (1) and (2) of Theorem 11.11. With the help of Lemma 14.7 (3) this will be easy.

14.17. *Proof of Theorem 11.11 (1), (2).* First, if (1) fails, we can find η such that the set of points $x \in A$ for which $\Theta_*^s(A \cap H(x, \theta, 2\eta), x) > 0$ for all $\theta \in S^{n-1}$ has positive $\mathcal{H}^s$ measure. Choose $\theta_1, \dots, \theta_k \in S^{n-1}$ such that for every $\theta \in S^{n-1}$ there is $i = 1, \dots, k$ with $|\theta - \theta_i| < \eta$; this implies for any $x \in \mathbf{R}^n$,

$$H(x, \theta_i, 2\eta) \subset H(x, \theta, \eta).$$

We can find positive numbers c and r_0 and a Borel subset B of A such that $\mathcal{H}^s(B) > 0$ and

$$\mathcal{H}^s(A \cap H(x, \theta_i, 2\eta) \cap B(x, r)) \geq cr^s$$

for $x \in B$, $0 < r < r_0$ and $i = 1, \dots, k$, whence

$$\mathcal{H}^s(A \cap H(x, \theta, \eta) \cap B(x, r)) \geq cr^s$$

for $x \in B$, $0 < r < r_0$, and for all $\theta \in S^{n-1}$. Since by Theorem 6.2 (1), $\Theta^{*s}(A, x) \leq 1$ for $\mathcal{H}^s$ almost all $x \in A$, this together with Remark 14.4 (3) implies that at $\mathcal{H}^s$ almost all $a \in B$ no tangent measure of $\mathcal{H}^s \llcorner A$ has support in a half-space with 0 on the boundary. Hence the negation of (1) leads to a contradiction with Lemma 14.7 (3).

If (2) fails, we can use (1) and Theorem 6.2 (1) to find a Borel subset B of A and positive numbers r_0 and c such that $\mathcal{H}^s(B) > 0$,

$$cr^s \leq \mathcal{H}^s(A \cap B(x, r)) \leq (3r)^s \quad \text{for } x \in B, \ 0 < r < r_0,$$

and that for all $x \in B$ there are sequences $\eta_i \downarrow 0$, $r_i \downarrow 0$ and $\theta_i \in S^{n-1}$ such that

$$\mathcal{H}^s(A \cap B(x, r_i) \cap H(x, \theta_i) \setminus H(x, \theta_i, \eta_i)) \geq cr_i^s$$

for $x \in B$. By passing to a sub-sequence we may assume $\theta_i \rightarrow \theta \in S^{n-1}$. It follows by Lemma 14.7 (1) and similar arguments as before that at $\mathcal{H}^s$ almost all points $a \in B$ some $\nu \in \text{Tan}(\mathcal{H}^s \llcorner A, a)$ satisfies the following.

There are positive numbers a and b and $\theta \in S^{n-1}$ such that

$$\begin{aligned} ar^s &\leq \nu(B(x, r)) \leq br^s \quad \text{for } x \in \text{spt } \nu, \ 0 < r < \infty, \\ \nu(\{x \in \mathbf{R}^n : x \cdot \theta = 0\}) &> 0. \end{aligned}$$

Since $s > n - 1$, this is impossible. □

As another application of Theorem 14.16 we have the following theorem. In Chapter 17 we shall see that much more is true.

14.18. Theorem. *Let m be an integer $0 \leq m \leq n$ and let μ be a Radon measure on $\mathbf{R}^n$ such that the density $\Theta^m(\mu, a)$ exists and is positive and finite at μ almost all points $a \in \mathbf{R}^n$. Then at μ almost all points $a \in \mathbf{R}^n$ there is $V \in G(n, m)$ such that $\mathcal{H}^m \llcorner V \in \text{Tan}(\mu, a)$.*

Proof. By Corollary 14.9 at μ almost all points the tangent measures of μ are m -uniform. Inspecting the proof of Theorem 14.10 we see that for $m < n$ the m -uniform measures possess tangent measures whose supports lie on a hyperplane. If $m = n - 1$, this completes the proof because of Theorem 14.16 (2) and Theorem 14.11. If $m < n - 1$ we continue to take new tangent measures, finally finding one whose support lies on an m -plane. $\square$

Remarks

14.19. Let K be a self-similar set with the open set condition, recall 4.13, and $\mu = \mathcal{H}^s \llcorner K$, where $s = \dim K$. Then (except for trivial cases such as when K is an interval) μ does not have a unique tangent measure at any points of K (the uniqueness of course means equality after multiplication by a constant). However, because of the self-similarity one would expect that the sceneries would fluctuate only in some restricted way when one approaches points $a \in K$, that is, $\text{Tan}(\mu, a)$ should not be too wild and it should not depend too strongly on a . Bedford and Fisher [2]–[3] proved results in this spirit for Cantor sets in $\mathbf{R}$; they showed that as sets the limits of the blow-ups are unique “up to $C^{1+\varepsilon}$ diffeomorphisms”. Bandt gave a tangent-measure-type formulation. He proved that if one averages suitably the measures $(r^{-s}T_{a,r}\mu) \llcorner B(a, r)$ over $[R, 1]$ with respect to the measure $r^{-1} dr$ and lets $R \downarrow 0$, then for μ almost all $a \in K$ one obtains in the limit a unique measure on the space of Radon measures on $B(1)$ which is independent of a , see Graf [3]. This is not so far away from the average densities of Bedford and Fisher, recall 6.14 (2); Graf [3] gave a precise connection. Another origin for Bandt’s tangential distribution can be found in the axiomatic theory for statistical self-similarity developed by U. Zähle [1]–[3], see in particular Patzschke and M. Zähle [2] where this becomes more visible.

Exercises.

1. Find all the tangent measures for $\mathcal{L}^n \llcorner B(1)$ and $\mathcal{H}^{n-1} \llcorner S^{n-1}$.
2. Prove the statements of Example 14.2 (3) in the case $m_k \rightarrow \infty$.
3. Construct a Radon measure μ which has a tangent measure ν such that $0 \notin \text{spt } \nu$.

4. Let $s > 0$ and suppose that there exist a Borel measure ν on $\mathbf{R}^n$ and positive numbers c and d such that $cr^s \leq \nu(B(x, r)) \leq dr^s$ for all $x \in \mathbf{R}^n$ and $0 < yr \leq 1$. Show that $s = n$.
5. Let ν be an s -uniform measure on $\mathbf{R}^n$. Show that for any non-negative Borel function $\varphi: \mathbf{R} \rightarrow \mathbf{R}$,

$$\int \varphi(|z - x|) d\nu z = \int \varphi(|z - y|) d\nu z \quad \text{for } x, y \in \text{spt } \nu.$$

6. Show that F_r defined in 14.12 is a metric on the space of Radon measures μ with $\text{spt } \mu \subset U(r)$.